A Simplified Model of Serpentine Locomotion

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Note. This document is an updated presentation of the dynamical modeling and control components of my 2017 MS thesis and paper [Racioppo and Ben-Tzvi, 2019].

Abstract

We revisit classical models of serpentine locomotion and show that, under appropriate assumptions, the dynamics admit a low-dimensional approximation governing the center-of-mass motion. While full models require solving constrained differential-algebraic systems, we demonstrate that, when the body follows a prescribed traveling-wave gait, the resulting motion can be approximated by a single first-order equation for the center-of-mass velocity. This reduction is obtained by combining a representative-link approximation with time-scale separation and period averaging. We identify the key mechanism underlying propulsion as a phase offset between lateral motion and body orientation, induced by anisotropic friction. In addition, we analyze the structure of the phase-offset optimization problem and derive a closed-form approximation for the optimal joint phase offset. Together, these results provide (i) a simplified dynamical model that predicts steady-state velocity, (ii) a principled characterization of optimal phase relationships, and (iii) an interpretation of serpentine motion in terms of traveling-wave structure and phase interactions. This perspective yields a unified and computationally efficient framework for understanding serpentine locomotion, connecting gait design, physical modeling, and control.

1 Introduction

Serpentine locomotion, or lateral undulation, is the dominant mode of motion in biological snakes and has long served as a template for snake robot design. It consists of a traveling transverse wave that propagates along the body. Unlike standing waves, this motion relies on a coordinated phase relationship between body segments: each segment pushes laterally against the ground while the overall wave advances from head to tail. When coupled with anisotropic friction—higher resistance normal to the body than along it—this lateral motion is converted into forward propulsion.

In robotic implementations, this continuous body deformation is approximated by a finite set of rigid links connected by actuated joints. The standard parameterization of planar serpentine motion prescribes the joint angles as a traveling sinusoidal wave:

$$\theta_{\mathrm{ref},i} = \alpha \sin(\omega t + (i-1)\delta) + \phi_0, \tag{1}$$

for $i \in \{1, \dots, N-1\}$. Here, α is the amplitude of angular displacement, ω is the temporal frequency, and δ is the phase offset between adjacent joints. The parameter ϕ_0 induces a constant curvature and enables turning.

While this representation is typically viewed as a high-dimensional control input over $N-1$ joints, it is in fact governed by a small set of parameters $(\alpha, \omega, \delta, \phi_0)$ that define a traveling wave along the body. In this sense, the full joint motion is determined by a low-dimensional parameterization of the waveform.

Despite this structure, most existing models of snake locomotion operate at the level of full multibody dynamics, requiring the solution of large systems of coupled differential-algebraic equations. While these models are physically accurate, they often make it difficult to isolate the mechanisms that govern locomotion performance, such as the role of phase relationships and anisotropic friction.

In this work, we develop two complementary perspectives on serpentine locomotion. First, we show that, when the body motion is prescribed as a traveling-wave gait and appropriate time-scale separation

holds, the center-of-mass dynamics can be approximated by a low-dimensional model. By treating the joint motion as a controlled input and averaging the resulting frictional forces, we derive a reduced description in which forward propulsion is governed by a phase offset between lateral motion and body orientation, and the resulting behavior is captured by a single first-order equation for the center-of-mass velocity.

Second, we analyze the structure of the phase-offset optimization problem and derive a closed-form approximation for the optimal joint phase offset. This analysis reveals how propulsion depends on a structured aggregation of phase interactions along the body, providing a complementary perspective on the role of the traveling-wave parameters.

By making these structures explicit, we aim to bridge the gap between detailed physical models and intuitive, interpretable descriptions of locomotion, providing tools that are both analytically tractable and physically grounded.

2 Related Work

Early work in snake robotics is largely attributed to Hirose [1993], who introduced the serpenoid curve as a mathematical description of serpentine locomotion. This formulation models the body of a snake as a traveling sinusoidal wave and remains the foundation for most subsequent work on snake robot kinematics.

Extensions of this idea have focused on alternative curvature parameterizations and their effect on locomotion efficiency. Ma [1999] analyzed different curvature profiles and compared their performance, while Chernousko [2005] proposed a complementary approach in which locomotion is constructed from elementary joint motions in low-degree-of-freedom systems. These works highlight that a wide range of locomotion patterns can be generated from relatively simple underlying structures.

A significant body of work has also focused on control and physical modeling. Liljebäck et al. [2012] and Liljebäck and Pettersen [2011] developed path-following and waypoint guidance controllers for snake robots executing serpenoid-like motions. In parallel, Liljebäck et al. [2010] established that anisotropic ground friction is a necessary condition for controllable serpentine locomotion, formalizing the role of directional frictional forces in generating propulsion. Further analysis of controllability and stability was provided by Liljebäck et al. [2011].

More comprehensive treatments of snake robot modeling and locomotion can be found in the survey by Transeth et al. [2009] and the monograph by Liljebäck et al. [2013], which cover both kinematic and dynamic approaches. Earlier experimental and control-oriented studies, such as Sato et al. [2002], also contributed to the development of practical locomotion strategies.

In addition to modeling and control, optimization of gait and design parameters has been widely studied, particularly with the goal of maximizing forward speed. This includes tuning link lengths, mass distributions, and control parameters to improve performance [Hirose, 1993, Chernousko, 2005].

Prior work has primarily approached serpentine locomotion from two complementary perspectives: kinematic descriptions based on traveling waves (e.g., the serpenoid curve), and high-fidelity dynamic models that capture frictional interactions and multibody effects. While both perspectives have led to significant advances, they often remain disconnected: kinematic models provide intuition but lack predictive power, whereas dynamic models are accurate but difficult to analyze and interpret.

This work aims to bridge this gap by making explicit the low-dimensional structure underlying serpentine locomotion. We show that the traveling-wave parameterization used in kinematic models induces a corresponding simplification in the dynamics, allowing the high-dimensional multibody system to be approximated by a single first-order equation governing center-of-mass motion. In this view, key quantities such as the phase offset and friction anisotropy emerge as the primary determinants of locomotion performance, providing a unified interpretation that connects gait design, physical modeling, and control.

3 Multibody Dynamic Model

We present a multibody dynamic (MBD) model of a snake robot, which serves as a high-fidelity reference for the simplified model developed in the subsequent section. Unlike reduced models that impose no-slip constraints, this formulation explicitly models frictional forces and internal constraint forces via Lagrange multipliers.

3.1 Kinematics and Constraints

The snake robot is modeled as N rigid links connected by revolute joints. Each link i has half-length L_i, mass m_i, and moment of inertia I_i. The generalized coordinates are:

$$q = [q_1, \ldots, q_N]^T, \quad q_i = [r_i, \phi_i]^T$$

where $r_i = [x_i, y_i]^T$ is the position of the center of mass and ϕ_i is the orientation.

Revolute joints impose $2(N-1)$ algebraic constraints, collected in:

$$\Phi(q, t) = 0.$$

The rotation matrix mapping global to local coordinates is:

$$R_i = \begin{bmatrix} \cos\phi_i & -\sin\phi_i \\ \sin\phi_i & \cos\phi_i \end{bmatrix}.$$

3.2 Friction Model

Friction is modeled using anisotropic coefficients c_n (normal) and c_t (tangential). The viscous friction force acting on link i is:

$$F_{f,i}(r_i, \phi_i) = -R_i^T \Omega_i R_i \dot{r}_i, \tag{2}$$

where $\Omega_i = \mathrm{diag}(c_{n,i}, c_{t,i})$.

Rotational friction is computed by approximating each half-link as n discrete masses with spacing:

$$l_{n,i} = \frac{2L_i}{2n+1}.$$

For angular velocity $\dot{\phi}_i$, segment velocities are:

$$v_{x_{n,i}} = \dot{\phi}_i l_{n,i} \cos\phi_i, \tag{3}$$

$$v_{y_{n,i}} = -\dot{\phi}_i l_{n,i} \sin\phi_i. \tag{4}$$

In the limit $n \to \infty$, this yields an equivalent force acting at:

$$l'_{\mathrm{eff},i} = \frac{1}{2}.$$

Defining:

$$v_{\mathrm{eff},i} = \begin{bmatrix} \dot{\phi}_i l'_{\mathrm{eff},i} \cos\phi_i \\ \dot{\phi}_i l'_{\mathrm{eff},i} \sin\phi_i \end{bmatrix},$$

the frictional torque is:

$$\tau_{f,i}(\phi_i) = l'_{\mathrm{eff},i} F_{f,i}(v_{\mathrm{eff},i}). \tag{5}$$

3.3 Equations of Motion

Let Q_A denote the generalized force vector and define:

$$M = \mathrm{diag}(m_1, m_1, I_1, \ldots, m_N, m_N, I_N).$$

The constrained variational equations are:

$$\delta q^T (M\ddot{q} - Q_A) = 0. \tag{6}$$

Introducing Lagrange multipliers λ, the system becomes the DAE:

$$\begin{bmatrix} M & \Phi_q^T & 0 \\ \Phi_q & 0 & 0 \\ 0 & 0 & I \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \\ \dot{q} \end{bmatrix} = \begin{bmatrix} Q_A \\ \gamma \\ \dot{q} \end{bmatrix}, \tag{7}$$

where Φ_q is the constraint Jacobian and γ contains acceleration-independent terms.

Initial conditions must satisfy:

$$\Phi_q \dot{q} = -\Phi_t.$$

The system state

$$Y = [\dot{q}^T \;\; \lambda^T \;\; q^T]^T$$

is obtained using a numerical DAE solver.

4 Partial Feedback Linearization of the Joint Dynamics

An N-link snake robot moving in the plane can be modeled using a multibody dynamic approach as a serial chain of straight links joined by revolute joints. The system's dynamics are described by $3N$ differential algebraic equations (DAEs) for the links, complemented by $2(N-1)$ algebraic constraint equations for the revolute joints.

The generalized coordinates for the i-th link's center of mass (CM) are specified by a vector $q_i = [r_i, \phi_i]^T$, where $r_i = [x_i, y_i]^T$ denotes the global position in the xy-plane, and ϕ_i is its global orientation. Introducing a vector of Lagrange multipliers $\lambda_i = [\lambda_{x,i}, \lambda_{y,i}]^T$ to enforce the joint constraints, a diagonal 3×3 generalized mass matrix M_i, a generalized external force vector $F_{\text{ext},i}$, and a vector of generalized actuation forces U_i for the i-th link, the system's equations of motion can be written as:

$$\begin{cases} M_i \ddot{q}_i = F_{\text{ext},i} + f_i(q, \dot{q}) + \lambda_i + U_i \\ \Phi_j(q, t) = 0 \end{cases} \tag{8}$$

Here, $f_i(q, \dot{q})$ represents internal forces and torques (e.g., Coriolis and centrifugal terms), while $\Phi_j(q, t) = 0$ represents the j-th revolute joint constraint equation. Since joint torques are the only control inputs, $U_i = [0, 0, \tau_i]^T$.

The goal of feedback linearization is to transform a nonlinear system into a linear one by canceling nonlinear terms with appropriate control inputs. For a snake robot, the angular dynamics are of primary interest. The control input can be decomposed as:

$$U_i = U_{\text{lin},i} + \bar{U}_i \tag{9}$$

The term $\bar{U}_i$ tracks a desired angular trajectory, while $U_{\text{lin},i}$ cancels nonlinear dynamics.

The full system can be written as:

$$M\ddot{q} = F_{\text{ext}}(q, \dot{q}) + F_{\text{int}}(q, \dot{q}) + \Phi_q^T \lambda + U \tag{10}$$

The angular component for link i is:

$$I_i \ddot{\phi}_i = (F_{\text{ext},i})_\phi + (F_{\text{int},i})_\phi + (\Phi_q^T \lambda)_i + \tau_i \tag{11}$$

Setting:

$$\tau_i = -(F_{\text{ext},i})_\phi - (F_{\text{int},i})_\phi - (\Phi_q^T \lambda)_i + \bar{U}_i$$

yields:

$$I_i \ddot{\phi}_i = \bar{U}_i \tag{12}$$

The DAE system is:

$$\begin{bmatrix} M & \Phi_q^T \\ \Phi_q & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Q_A \\ \gamma \end{bmatrix} \tag{13}$$

Expanding:

$$M\ddot{q} + \Phi_q^T \lambda = Q_A \tag{14}$$

$$\Phi_q \ddot{q} = \gamma \tag{15}$$

Solving:

$$\ddot{q} = M^{-1}(Q_A - \Phi_q^T \lambda) \tag{16}$$

$$\Phi_q M^{-1}(Q_A - \Phi_q^T \lambda) = \gamma \tag{17}$$

$$\Phi_q M^{-1} Q_A - \Phi_q M^{-1} \Phi_q^T \lambda = \gamma \tag{18}$$

Thus:

$$\Phi_q M^{-1} \Phi_q^T \lambda = \Phi_q M^{-1} Q_A - \gamma \tag{19}$$

Let $A = \Phi_q M^{-1} \Phi_q^T$. Then:

$$\lambda = A^{-1}(\Phi_q M^{-1} Q_A - \gamma) \tag{20}$$

Substituting λ back yields the linearizing control.

A key feature of this formulation is that feedback linearization is performed directly on the differential-algebraic system, rather than after reducing it to an ODE via coordinate elimination. By retaining the constraint equations and explicitly solving for the Lagrange multipliers λ, the controller has direct access to the internal constraint forces.

This makes it possible to cancel the contribution of $(\Phi_q^T \lambda)$ within the control law, rendering the angular acceleration of each joint locally decoupled from the rest of the chain.

Working in the DAE framework also avoids the need to form and manipulate the full joint-space mass matrix, whose complexity grows rapidly with the number of links. Instead, the structure of the constraint equations is exploited directly, yielding a computationally efficient and exact linearization of the angular dynamics.

4.1 Traveling-Wave Control Law

Under the feedback-linearized dynamics, the joint angles can be driven to track a prescribed trajectory. In this work, we consider the standard traveling-wave parameterization:

$$\theta_{\mathrm{ref},i} = \alpha \sin(\omega t + (i-1)\delta) + \phi_0.$$

This control law enforces a coordinated wave of curvature along the body. The phase offset δ determines the spatial structure of the wave, while ω sets its temporal frequency. As shown in the following section, the interaction between this imposed kinematic pattern and anisotropic friction gives rise to forward propulsion. Forward propulsion is governed primarily by the phase offset δ in combination with anisotropic friction. Setting $\phi_0 = 0$ produces straight-line motion, while nonzero ϕ_0 induces turning by introducing a constant curvature.

5 Optimal Phase Offset and Traveling-Wave Structure

In this section, we analyze the structure of the phase-offset optimization problem introduced in Liljebäck et al. [2013]. Starting from the original double-sum formulation, we show that it can be reduced to a single weighted sum over phase differences along the body.

This reduction reveals that the objective depends only on relative phase differences, with weights that vary systematically along the chain. The problem of maximizing propulsion can therefore be understood as optimizing a structured weighted sum of sinusoidal phase differences, rather than an arbitrary high-dimensional function.

5.1 Background and Problem Formulation

Following Liljebäck et al. [2013], the phase offset that maximizes forward velocity can be approximated as the maximizer of:

$$k_\delta = \sum_{i=1}^{N-1} \sum_{j=1}^{N-1} c_{ij} \sin((j-i)\delta) \tag{21}$$

where $C = AD^T(DD^T)^{-1}$, and

$$A = \begin{bmatrix} 1 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 1 & -1 \end{bmatrix}.$$

This expression encodes how phase differences between joints contribute to propulsion.

The double summation in k_δ obscures the underlying structure of the objective. To obtain a tractable expression, we exploit the banded structure of D and compute how contributions accumulate along its diagonals. This allows the double sum to be reduced to a single weighted summation over phase differences.

5.2 Reduction to a Single Summation

Let $n = N - 1$ and define $B = (DD^T)^{-1}$. Since DD^T is symmetric, B is also symmetric. We compute the diagonal sums of $B = (DD^T)^{-1}$ and propagate this structure through $C = AD^TB$, which allows the double sum to collapse into a single weighted summation.

Proposition 1. Let $B = (DD^T)^{-1} \in \mathbb{R}^{n\times n}$. The sum of the k-th diagonal of B is

$$b_k = \frac{T_k}{n+1}, \quad \text{where} \quad T_k = \frac{k(k+1)(k+2)}{6}.$$

Here, the k-th diagonal is indexed from the outermost diagonal (bottom-left / top-right) inward toward the main diagonal.

Proof. We establish three properties of B.

(1.1) The entries along the boundary of B increase linearly in steps of $\frac{1}{n+1}$.

(1.2) The first n diagonals of B are recursively generated from the bottom row and scale with $\frac{1}{n+1}$.

(1.3) The sum of the main diagonal is $b_n = \frac{T_n}{n+1}$.

Proof of (1.1). Consider the first column of B. The defining relation $DD^TB = I$ implies:

$$-b_{n,1} + 2b_{n-1,1} - b_{n-2,1} = 0 \quad \Rightarrow \quad b_{n-1,1} = 2b_{n,1},$$

$$-b_{n-2,1} + 2b_{n-1,1} - b_{n-3,1} = 0 \quad \Rightarrow \quad b_{n-2,1} = 3b_{n,1}.$$

Proceeding inductively, suppose:

$$b_{n-i,1} = (i+1)b_{n,1}, \quad i \geq 2.$$

Then:

$$-b_{n-(i+1),1} + 2b_{n-i,1} - b_{n-(i-1),1} = 0$$

implies:

$$b_{n-(i+1),1} = 2(i+1)b_{n,1} - ib_{n,1} = (i+2)b_{n,1}.$$

Thus, the first column increases linearly. At the top:

$$b_{1,1} = n\, b_{n,1}.$$

Since $b_{n,1} = \frac{1}{n+1}$, we obtain:

$$b_{1,1} = \frac{n}{n+1}.$$

By symmetry, the same structure holds for the last column, and therefore for the top and bottom rows.

Proof of (1.2). Consider the i-th element of the k-th diagonal:

$$b_{n-k-(i-1),\, i+k-1}, \quad k \in [1, n],\ i \in [1, k].$$

These entries satisfy the recurrence:

$$-b_{n-k-(i-1),\, i+k-1} + 2b_{n-k-i,\, i+k} - b_{n-k-(i+1),\, i+k+1} = 0,$$

with boundary condition:

$$b_{n-k-(i-1),\, i+k-1} = 2b_{n-k-i,\, i+k}$$

at the second-to-bottom row.

Thus, each diagonal is generated recursively from the bottom row, and all diagonals share the same structure up to the scaling factor $\frac{1}{n+1}$.

Proof of (1.3). The matrix

$$DD^T = \begin{bmatrix} 2 & -1 & & 0 \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ 0 & & -1 & 2 \end{bmatrix}$$

is the second-difference matrix.

Using $DD^T B = I$ and taking traces:

$$\mathrm{tr}(DD^T B) = n.$$

This yields the identity:

$$\sum_{i=1}^{n-1} (b_{i,i} - b_{i,i+1}) + b_{n,n} = 1.$$

Let $f(k)$ denote the sum of the k-th diagonal (counted from the outside inward). Then the recursion:

$$f_n = 2f_{n-1} - f_{n-2} + \frac{n(n-1)}{2}$$

matches the tetrahedral sequence:

$$f_n = \frac{T_n}{n+1}.$$

Conclusion. By induction on n, all diagonal sums satisfy:

$$b_k = \frac{T_k}{n+1},$$

which proves Proposition 1.

□

Proposition 2. The sum of the k-th diagonal of $C = AD^T(DD^T)^{-1}$ is:

$$c_k = \pm \frac{2t_k}{n+1}, \quad t_k = \frac{k(k+1)}{2},$$

where the sign is positive in the upper triangle and negative in the lower triangle.

Proof. Multiplication by D^T subtracts adjacent diagonals:

$$\sum_i b_{i,k+i} \mapsto \sum_i (b_{i,k+i} - b_{i,k+i-1}).$$

Thus, diagonal sums become:

$$t_k - t_{k-1} = t_k,$$

where t_k are triangular numbers.

Multiplication by A adds adjacent rows. Equivalently, each diagonal is added to the one above it, doubling its contribution (except for boundary corrections).

Accounting for boundary terms, we obtain:

$$c_k = \pm \frac{2t_k}{n+1}.$$

□

Proposition 3. The objective function simplifies to:

$$k_\delta = \sum_{k=1}^{N-1} \frac{2k(k+1)}{N} \sin((N-1-k)\delta). \tag{22}$$

Proof. Since each diagonal corresponds to $j - i = k$, the sine term is constant along each diagonal. Using the result of Proposition 2:

$$k_\delta = \sum_{k=1}^{N-1} \left(\frac{2t_k}{n+1} \sin(k\delta) - \frac{2t_k}{n+1} \sin(-k\delta) \right).$$

$$= \frac{2}{n+1} \sum_{k=1}^{N-1} t_k \sin(k\delta) + \frac{2}{n+1} \sum_{k=1}^{N-1} t_k \sin(k\delta).$$

Substituting $t_k = k(k+1)/2$ and $n = N - 1$ yields:

$$k_\delta = \sum_{k=1}^{N-1} \frac{2k(k+1)}{N} \sin((N-1-k)\delta).$$

□

This reduction shows that the objective depends only on phase differences between links, with weights that increase along the body. The contribution of each phase difference is therefore not uniform, but follows a structured weighting induced by the underlying model.

5.3 Closed-Form Approximation

To obtain an explicit approximation for the optimal phase offset, we differentiate the reduced objective function

$$k_\delta = \sum_{k=1}^{N-1} \frac{2k(k+1)}{N} \sin((N-1-k)\delta)$$

and set the derivative equal to zero.

This yields a nonlinear equation in $\bar{\delta}$ of the form:

$$1 + \frac{N(N-1)}{N+1} \cos(\bar{\delta}) + \frac{N-2}{N+1} \cos(\bar{\delta} N) = \cos(\bar{\delta}(N-1))^{N-2}. \tag{23}$$

To obtain a tractable expression, we perform a Taylor expansion about $\bar{\delta} = 0$ and retain terms up to fourth order. We then expand the resulting expression for large N and keep the leading terms.

This yields the approximation:

$$\delta_{\text{opt}} \approx \frac{4.10}{N} + \frac{2.15}{N^2}. \tag{24}$$

We can also write this in terms of the number of joints as:

$$\delta_{\text{opt}} \approx \frac{4.10}{N-1} - \frac{1.95}{(N-1)^2} \tag{25}$$

This expression approximates the original formulation defined by k_δ to within approximately 2.75% error over the range $3 \leq N \leq 100$.

5.4 Limiting Behavior

The above approximation admits a simple interpretation in both small- and large-N regimes.

For $N = 3$ (the smallest possible number of links), the reduced objective simplifies to

$$k_\delta = \frac{4}{3}\sin(\delta),$$

which is maximized at

$$\delta_{\text{opt}} = \frac{\pi}{2} = \frac{\pi}{N-1}.$$

Thus, in the minimal case of two joints, the system reduces to a single phase interaction with a sinusoidal optimum.

As $N \to \infty$, the higher-order term becomes negligible, and the optimal phase offset scales as

$$\delta_{\text{opt}} \sim \frac{4.10}{N-1}.$$

Interpretation. Hence, the optimal phase offset per joint increases from π at $N = 3$ to approximately 4.10 as $N \to \infty$. This characterizes a transition from a low-dimensional regime, in which the behavior is governed by a single phase interaction, to a many-body regime in which the optimal phase offset is determined by the cumulative effect of weighted interactions along the body.

6 Reduced-Order Model of Lateral Undulation

We develop a reduced-order model of serpentine locomotion by combining a representative-link approximation with time-scale separation and period averaging. Rather than solving the full multibody dynamics, we analyze the forces generated by a prescribed traveling-wave motion and derive their effect on the slower evolution of the center-of-mass velocity. Under these assumptions, the system admits a low-dimensional description that captures the leading-order mechanism of propulsion.

6.1 Setup and Friction Model

Consider a snake with N links of uniform length L and mass m undergoing planar motion with link centers $r_i = [x_i \quad y_i]^T$. Assuming motion primarily in the y-direction, the acceleration of link i under anisotropic viscous friction is:

$$\ddot{r}_i = -\frac{1}{m} R_i^T \Omega_i R_i \dot{r}_i \tag{26}$$

where $\Omega_i = \text{diag}(c_{n,i}, c_{t,i})$ and R_i is the rotation matrix associated with orientation

$$\phi_i = a\sin(\omega t + (i-1)\delta).$$

Expanding the friction term:

$$-R_i^T \Omega_i R_i = -\begin{bmatrix} c_n \cos^2\phi_i + c_t \sin^2\phi_i & (c_n - c_t)\sin\phi_i\cos\phi_i \\ (c_n - c_t)\sin\phi_i\cos\phi_i & c_n \sin^2\phi_i + c_t\cos^2\phi_i \end{bmatrix}.$$

The diagonal terms correspond to drag, while the off-diagonal terms couple lateral velocity and orientation, and are responsible for propulsion.

6.2 Assumptions and Reduction Principle

We consider the regime:

- Low period-averaged acceleration (near steady-state motion)
- Small amplitude $a \ll 1$
- Phase offset $\delta \approx \bar{\delta}$ (near optimal)

The sinusoidal form of ϕ_i is imposed by the control law, which drives the joints to track a traveling-wave trajectory. As a result, the body shape is not determined by passive dynamics, but is actively enforced up to tracking error.

The analysis therefore does not attempt to solve the full coupled dynamics. Instead, it evaluates the forces generated by this prescribed gait and determines their effect on the slower evolution of the center-of-mass motion. This is justified by two approximations.

First, under the feedback-linearized control law described in the previous section, the angular dynamics are input-dominated: the controller cancels nonlinear and constraint-induced forces, allowing the joint angles to track the prescribed traveling-wave trajectory with small error. As a result, the angular motion is well approximated by

$$\phi_i(t) \approx \alpha \sin(\omega t + (i-1)\delta),$$

and its steady-state amplitude is determined primarily by the control input.

Second, the system exhibits a separation of time scales. The joint oscillations occur at frequency ω, while the forward velocity evolves under the cumulative effect of friction forces, which are $O(a^2)$ for small amplitude. Thus, the center-of-mass motion varies more slowly than the joint motion, and averaging over one oscillation period is justified.

Finally, since each link executes the same waveform up to a phase shift, the body samples all phases of the traveling wave. As a result, the total propulsive force can be approximated, to leading order, by averaging the contribution of a representative link over one oscillation period and scaling by the number of links. This approximation neglects spatial variations in force contribution along the body, which enter as higher-order effects.

Under these conditions, the coupling from translational dynamics back into the angular subsystem enters only at higher order in the amplitude. Consequently, we adopt a one-way coupling approximation: the imposed angular motion drives the translational dynamics, while the feedback from translation to joint motion is neglected to leading order. This yields a reduced model with hierarchical coupling, in which the steady-state amplitude $\bar{a}$ is determined by the angular control law and treated as an input to the translational dynamics.

6.3 Kinematics of a Representative Link

Let link k satisfy $\phi_k = 0$ at $t = 0$, so that:

$$\phi_k = a \sin(\omega t).$$

Here, a denotes the steady-state oscillation amplitude induced by the control input α.

To determine its lateral motion, we approximate the body as closely following the serpenoid curve and sum the projections of neighboring links onto the x-axis. The maximum lateral displacement occurs halfway between consecutive zero-orientation points. Enforcing symmetry gives the constraint:

$$\sum_{i=0}^{2m} \sin(i\delta) = 0 \quad \Rightarrow \quad m = \frac{\pi}{\delta}.$$

Thus, the lateral displacement of link k is approximated by:

$$x_k \approx -\frac{L}{2} \sum_{i=1}^{\pi/\delta} \sin\big(a \sin(\omega t + i\delta)\big).$$

For $a \ll 1$, we linearize $\sin(a\sin(\cdot)) \approx a\sin(\cdot)$:

$$x_k \approx -\frac{La}{2}\sum_{i=1}^{\pi/\delta}\sin(\omega t + i\delta).$$

Expanding the sum and differentiating yields:

$$\dot{x}_k \approx \frac{La\omega}{2}\csc\left(\frac{\delta}{2}\right)\sin\left(\omega t + \frac{\delta}{2}\right). \tag{27}$$

This result shows that the lateral velocity is phase-shifted by $\delta/2$ relative to the orientation $\phi_k = a\sin(\omega t)$. This phase shift arises geometrically from the spatial structure of the traveling wave.

6.4 Propulsive Force Generation

The y-component of the friction force is:

$$F_{ky} = -\left[(c_n - c_t)\sin\phi_k\cos\phi_k\dot{x}_k + (c_n\sin^2\phi_k + c_t\cos^2\phi_k)\dot{y}_k\right].$$

The first term produces propulsion. Substituting $\phi_k = a\sin(\omega t)$ and the expression for $\dot{x}_k$, and using small-angle approximations:

$$\sin\phi_k \approx a\sin(\omega t), \quad \cos\phi_k \approx 1,$$

gives:

$$F_{ky,\text{prop}} \approx -\frac{La^2\omega(c_n - c_t)}{2}\csc\left(\frac{\delta}{2}\right)\sin(\omega t)\sin\left(\omega t + \frac{\delta}{2}\right).$$

Thus, propulsion arises from the interaction of two oscillatory signals with a phase offset of $\delta/2$. Averaging over one period yields a nonzero mean force proportional to the sine of the phase difference between these signals. This nonzero average is the source of net forward propulsion.

6.5 Angular Dynamics and Control

The angular dynamics of link k are:

$$I\ddot{\phi} = \tau_u - f_v\dot{\phi} + f_i(\phi, \dot{\phi}),$$

where

$$f_v = \frac{2}{3}c_n\left(\frac{L}{2}\right)^2.$$

Using PD control:

$$\tau_u = k_p[\alpha\sin(\omega t) - \phi] + k_d[\alpha\omega\cos(\omega t) - \dot{\phi}].$$

Assuming cancellation of nonlinear terms:

$$I\ddot{\phi} + c\dot{\phi} + b\phi = k_p\alpha\sin(\omega t) + k_d\alpha\omega\cos(\omega t).$$

Solving for steady-state response gives:

$$\bar{a} = \frac{\alpha\sqrt{k_p^2 + k_d^2\omega^2}}{\sqrt{(k_p - I\omega^2)^2 + (k_d\omega)^2}} \tag{28}$$

This steady-state amplitude $\bar{a}$ is computed under the assumption that the angular dynamics are dominated by the control input, with only higher-order influence from the translational motion.

6.6 Reduction to a Single Dynamical Equation

Using the one-way coupling approximation described above, we treat $\bar{a}$ as determined by the angular control law and independent of the forward velocity to leading order.

Under the above assumptions, the forward dynamics evolve on a slower time scale than the joint oscillations, and the total propulsive force is approximated, to leading order, by the averaged contribution of a representative link. This permits reduction of the full $3N$-dimensional system to a single equation governing the center-of-mass velocity $Y = \dot{y}$:

$$\dot{Y}(t) = f_x(t) - f_y(t)Y(t), \tag{29}$$

where

$$\begin{cases} f_x(t) = C_1 \bar{a}\omega \sin\bar{\phi}\cos\bar{\phi}\sin\left(\omega t + \frac{\bar{\delta}}{2}\right) \\ f_y(t) = C_2 \sin^2\bar{\phi}. \end{cases} \tag{30}$$

Here, $f_x(t)$ captures propulsion arising from phase interaction, while $f_y(t)$ captures orientation-dependent drag.

6.7 Interpretation

Forward motion arises from a phase offset between lateral velocity and body orientation. If these signals are in phase, the motion is reciprocal over a cycle and produces no net displacement. A phase offset—induced by the traveling-wave structure—breaks this symmetry, and in the presence of anisotropic friction generates net propulsion.

The reduced model yields several scaling relationships. The steady-state velocity satisfies $Y_{\text{ss}} \propto \bar{a}^2\omega(c_n - c_t)$, exhibiting quadratic dependence on oscillation amplitude, linear dependence on driving frequency, and requiring anisotropic friction ($c_n > c_t$) for propulsion.

The reduced model also enables analysis of the optimal driving frequency. Since the steady-state velocity scales as $Y_{\text{ss}} \propto \bar{a}^2(\omega)\,\omega$, maximizing propulsion requires balancing the frequency-dependent amplitude response of the joint dynamics with the linear increase in propulsion with ω. Using the expression for $\bar{a}$, this leads to an implicit optimality condition for ω, which can be solved numerically. In the lightly damped regime, the optimal frequency occurs near the natural frequency of the joint dynamics, $\omega_{\text{opt}} \approx \sqrt{k_p/I}$, with damping shifting the optimum to lower values. This highlights that efficient locomotion arises from tuning the driving frequency to the mechanical response of the system.

The model also provides a characteristic convergence time scale. Linearizing the dynamics gives $\tau \sim 1/\langle f_y\rangle \propto 1/\bar{a}^2$, indicating that larger amplitudes lead to faster convergence to steady-state motion.

Finally, the reduced model predicts a dependence on phase offset of the form $Y_{\text{ss}} \propto \cos(\delta/2)$. Combining this with the scaling $\delta \sim 1/N$ from the phase analysis in Sec. 5 implies that increasing the number of links increases propulsion efficiency, with the steady-state speed approaching a finite limit as $N \to \infty$. For example, comparing $N = 3$ (where $\delta \approx \pi/2$) with the large-N limit gives an increase in speed by a factor of $\sqrt{2}$. However, because the reduced model neglects spatial variation in force contributions along the body, it does not capture the finite optimal phase offset, which arises from distributed interactions in the full system.

7 Path Following

To evaluate the feedback-linearized controller in a control setting, we consider a path-following task in which the snake robot tracks a sequence of waypoints by modulating the gait offset parameter ϕ_0.

Under the traveling-wave control law (Equation (8)), the robot moves in a straight line when $\phi_0 = 0$ and follows a trajectory with approximately constant curvature when ϕ_0 is held constant [Sato et al., 2002]. Thus, steering can be achieved by dynamically adjusting ϕ_0.

A common approach [Liljebäck et al., 2013] defines the reference offset as:

$$\phi_{0,\text{ref}} = A_h(h_{\text{ref}} - h), \tag{31}$$

where h is the average heading of the robot,

$$h = \frac{1}{N-1} \sum_{i=1}^{N-1} \theta_i,$$

and h_{ref} is the desired heading direction.

For waypoint tracking, the desired heading can be computed geometrically as:

$$h_{\text{ref}} = \arctan\left(\frac{P_Y}{\Delta - P_X}\right), \tag{32}$$

where (P_X, P_Y) represents the relative position of the target waypoint.

To improve robustness, we instead formulate heading control in terms of cross-track error. Let $K_{i,j}$ denote the position of the robot and E_j the j-th waypoint. The cross-track error is defined as:

$$P_{X,j} = \pm\sqrt{(K_{i,j,x} - E_{j,x})^2 + (K_{i,j,y} - E_{j,y})^2}, \tag{33}$$

where the sign is determined by the relative orientation between the robot velocity and the waypoint direction:

$$P_{X,j} = -\text{sgn}(\theta_{\text{rel}} - \theta_{\text{wp}})\, P_{X,j}.$$

Here,

$$\theta_{\text{rel}} = \arctan 2(v_{i,j,y}, v_{i,j,x}), \quad \theta_{\text{wp}} = \arctan 2(w_{j,y}, w_{j,x}), \tag{34}$$

represent the robot's instantaneous heading and the direction to the waypoint, respectively.

The desired heading is then generated using a PID controller:

$$h_{\text{ref},j} = K_P P_{X,j} + K_I \int P_{X,j}\, dt + K_D \dot{P}_{X,j}. \tag{35}$$

To improve performance across different motion regimes, the proportional gain may be adapted based on the angle of attack ϕ_{AoA}:

$$K_P = p_1\big(\arctan 2(p_2 \phi_{\text{AoA}}) + p_3\big). \tag{36}$$

This control strategy modulates the gait offset ϕ_0 to steer the robot while the feedback-linearized controller ensures accurate tracking of the underlying traveling-wave motion.

8 Experiments

The reduced model was compared to a full multibody simulation using MATLAB `ode45` over a 30-second interval, with the system initialized from rest.

At fixed parameter settings, the reduced model reproduces the key qualitative features of the multibody dynamics. In particular, both models exhibit a transient oscillatory regime in the center-of-mass velocity and convergence to a steady-state forward speed, as shown in Figure 1. The dependence of both oscillation amplitude and steady-state velocity on the driving frequency ω is also captured. Across a range of frequencies, higher ω leads to smaller oscillation amplitudes and higher steady-state speeds, a trend that is evident in Figure 1.

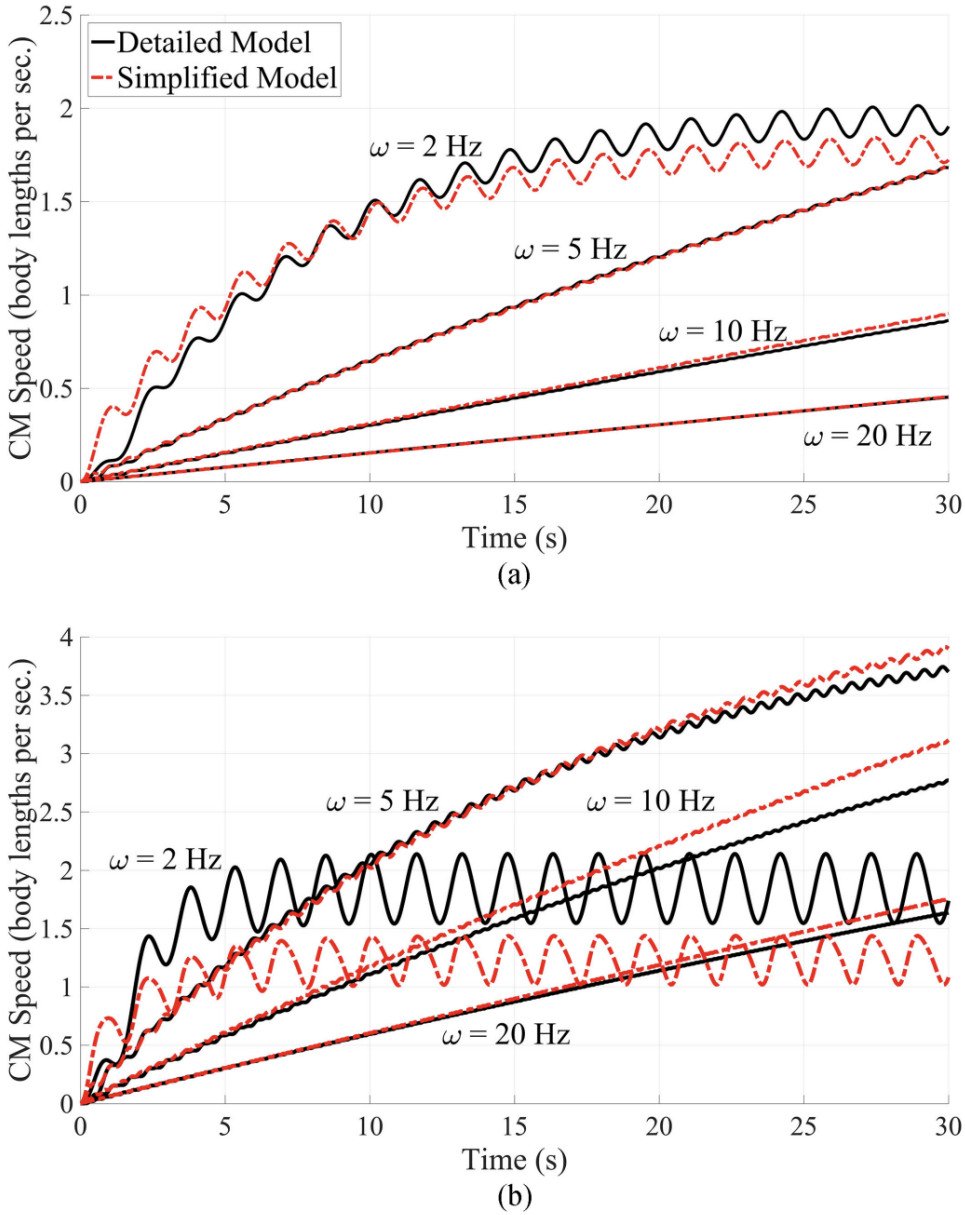

Figure 1: Comparison of the full multibody dynamic model (black solid line) with the single-body model (red dotted-line) at various oscillation frequencies for (a) $\alpha = 0.5$ rad and (b) $\alpha = 1$ rad.

Quantitatively, the agreement depends on the operating regime. For sufficiently large ω, the reduced model achieves $< 1\%$ error in steady-state amplitude for $\omega > 10$ Hz, while at lower frequencies the error increases to $\sim 40\%$ at $\omega = 2$ Hz. This degradation is consistent with the increasing discrepancy between the two models visible at lower frequencies in Figure 1.

These discrepancies follow directly from the assumptions underlying the reduced model. The derivation relies on (i) small-amplitude motion, (ii) near steady-state dynamics, and (iii) a separation of time scales between fast joint oscillations and the slower evolution of the center-of-mass velocity. At low frequencies, this time-scale separation breaks down, and the forward velocity $Y(t)$ no longer evolves slowly relative to the oscillatory forcing terms, invalidating the averaging argument. Similarly, at larger amplitudes, the small-angle approximations used in deriving $\dot{x}_k$ and the propulsive force no longer hold, and inter-link coupling effects become significant.

In addition, the reduced model collapses the full multibody dynamics into a small number of effective coefficients (e.g., C_1, C_2), which absorb the effects of constraint forces and coupling. While this captures the dominant behavior when motion is smooth and coherent, it becomes sensitive to parameter choices when these conditions are violated.

The dependence of steady-state speed on frequency and body length is shown in Figure 2. Increasing frequency leads to higher steady-state speeds, while longer bodies produce greater propulsion due to increased phase-coherent interactions along the chain.

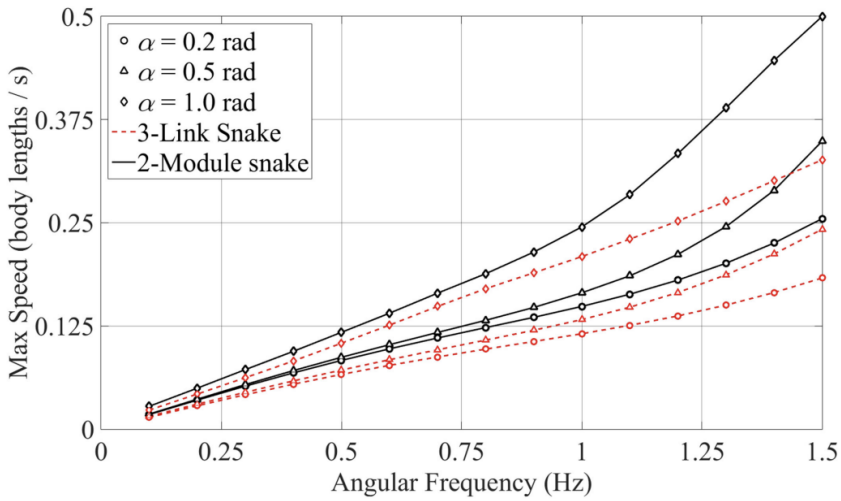

Figure 2: Simulated steady-state speeds for snake robots with different body lengths as the driving amplitude and frequency are varied.

Overall, the reduced model should be interpreted as capturing the dominant dynamical structure of serpentine locomotion rather than providing uniformly accurate predictions. Within its regime of validity, it offers a compact description of how phase relationships and anisotropic friction generate propulsion.

Finally, Figure 3 illustrates a path-following task using the proposed control strategy. By modulating the offset parameter ϕ_0, the snake robot is able to track a sequence of waypoints and execute complex planar trajectories.

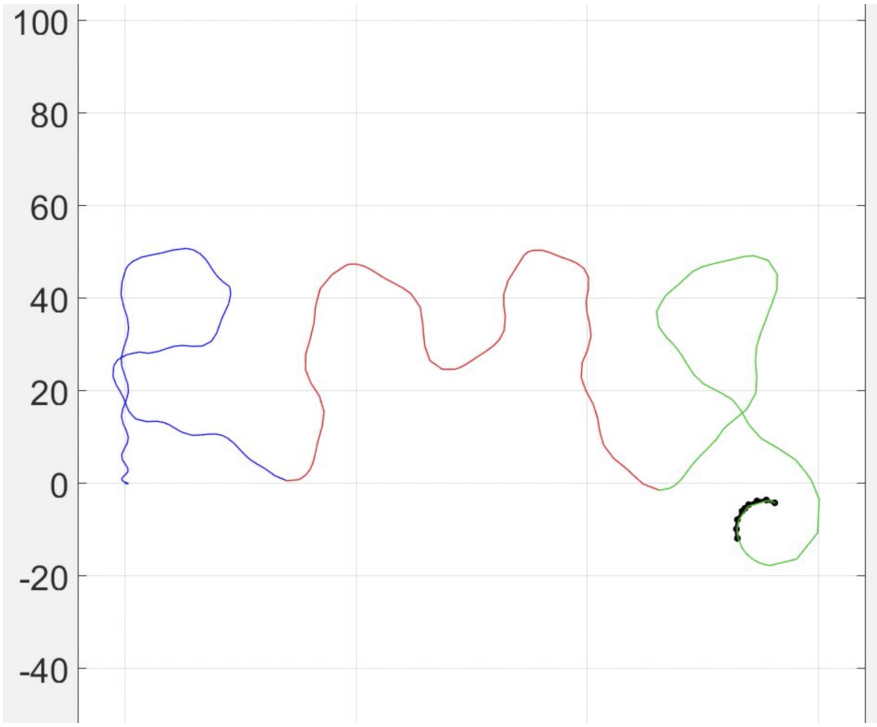

Figure 3: Simulated path-following control using waypoint guidance.

9 Conclusion

We revisited classical models of serpentine locomotion and showed that, despite the apparent complexity of multibody dynamics, the system admits a simple underlying structure. In particular, we demonstrated that locomotion can be understood as a low-dimensional traveling-wave process, in which both kinematics and dynamics are governed by a small set of parameters.

By analyzing the interaction between body orientation, lateral motion, and anisotropic friction, we derived a reduced dynamical model in which forward velocity is governed by a single first-order equation. This formulation makes explicit the mechanism of propulsion as a phase offset between lateral velocity and body orientation, induced by the traveling-wave structure of the gait.

We further showed that the optimal joint phase offset can be approximated in closed form and is determined by a structured weighting of phase interactions along the body, rather than a purely geometric phase partition. This provides a principled connection between gait parameters and locomotion performance.

Together, these results unify kinematic descriptions, dynamic models, and control strategies within a common low-dimensional framework. Beyond computational efficiency, this perspective provides an interpretable account of how coordinated body motion and anisotropic friction generate propulsion, and may inform the design of more effective control policies and robot morphologies.

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